Introduction to Statistical Learning and Application - Master of Applied Mathematics

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March 5, 2026

Outline

1. Classification
2. Logistic Regression
3. SVM
4. Evaluation metrics
5. Lab: Classifcation Methods

Motivation

The linear regression model assumes that the response variable Y is quantitative.

- Predicting a qualitative response for an observation can be referred to as **classifying** that observation, since it involves assigning the observation to a category, or class.
- On the other hand, often the methods used for classification first **predict the probability** that the observation belongs to each of the categories of a qualitative variablen.

Why not regression?

1. A regression method cannot accommodate a qualitative response with more than two classes;
2. A regression method will not provide meaningful estimates of $Pr(Y|X)$, even with just two classes.

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What is Classification?

Definition

Classification is a type of supervised machine learning where the goal is to categorize input data into predefined classes or labels.

Types of Classification Algorithms

1. Linear Classifiers:
   - Logistic Regression
   - Support Vector Machines (SVM)
2. Nearest Neighbor Classifiers:
   - k-Nearest Neighbors (k-NN)
3. Tree-Based Classifiers:
   - Decision Trees
   - Gradient Boosting Machines (e.g., XGBoost)
4. Naive Bayes Classifiers

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   - The Logistic Model
   - Estimating the Regression Coefficients
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The `Default` dataset (credit risk example)

We study a binary classification problem: default $\in \{\text{Yes}, \text{No}\}$
The goal is to predict whether an individual will default on their credit card payment.

- Default: missing a required credit card payment
- Balance: average monthly *credit card debt* (not bank money)
- Income: annual income of the individual

Key observations

- High balances are strongly associated with default
- Income alone is a weak predictor
- Classes overlap $\Rightarrow$ no perfect separation

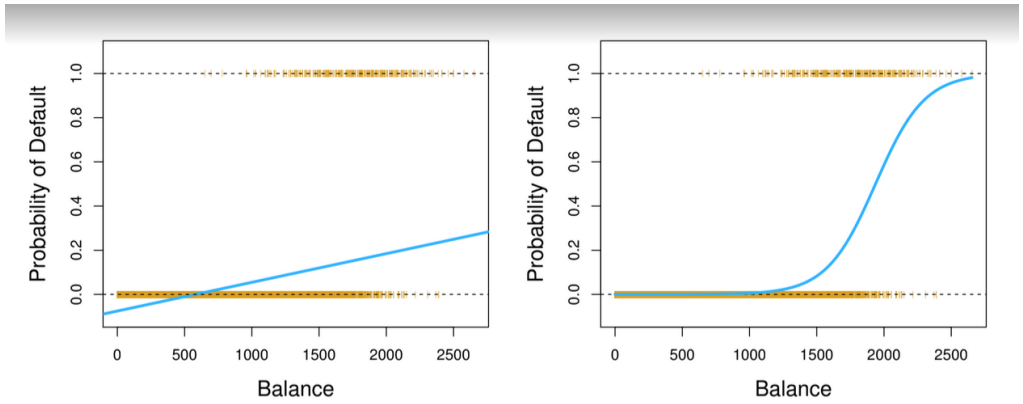
Logistic Regression

Key Note

Rather than modeling this response Y directly, logistic regression models the probability that Y belongs to a particular category.

$$p(X) = \beta_0 + \beta_1(X) \tag{1}$$

Example: For the Default data, we model Pr(default = Yes|balance).



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5 Lab: Classifcation Methods

The Logistic Model

Problem

For balances close to zero, we predict a negative probability of default; if we were to predict for very large balances, we would get values bigger than 1.

To avoid this problem, we must model p(X) using a function that gives outputs between 0 and 1 for all values of X. Many functions meet this description. In logistic regression, we use the logistic function,

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}} \tag{2}$$

The Logistic Model

- After a bit of manipulation of (2), we find that:

$$\frac{p(X)}{1-p(X)} = e^{\beta_0+\beta_1 X} \tag{3}$$

- The left-hand side is called the odds.
- By taking the logarithm of both sides of (3), we arrive at

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X.$$

- The left-hand side is called the log odds or logit.

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4 Evaluation metrics

5 Lab: Classifcation Methods

Estimating the Regression Coefficients- The method of maximum likelihood

The basic intuition behind using maximum likelihood

We seek estimates for β_0 and β_1 such that the predicted probability $\hat{P(xi)}$ of default for each individual, using (2), corresponds as closely as possible to the individual's observed default status.

This intuition can be formalized using a mathematical equation called a likelihood function:

$$\ell(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i':y_{i'}=0} (1 - p(x_{i'})).$$

Why Maximum Likelihood for Logistic Regression?

Key idea (high level)

In logistic regression, we model probabilities of a binary outcome, not the outcome itself.

- In linear regression: coefficients are estimated by least squares (closed form).
- In logistic regression: $y_i \in \{0, 1\}$ and $\hat{p}_i$ must be in $[0, 1]$.
- We choose coefficients β that make the **observed labels** most plausible under the model.

$$p_i = \mathbb{P}(Y_i = 1 \mid x_i) = \sigma(\eta_i), \quad \eta_i = \beta_0 + \beta^\top x_i, \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

From Individual Probabilities to a Likelihood

- For each observation i:
  - if $y_i = 1$, the model assigns probability p_i
  - if $y_i = 0$, the model assigns probability $1 - p_i$
- The likelihood multiplies contributions across all observations.

Likelihood of the full sample

$$L(\beta) = \prod_{i=1}^{n} p_i^{y_i} (1 - p_i)^{1 - y_i}$$

- Maximizing $L(\beta)$ means: predicted probabilities match observed outcomes as well as possible.

Why Use the Log-Likelihood?

- Products of many probabilities can be numerically unstable.
- Taking logs converts products into sums (easier to optimize).
- Maximizing $L(\beta)$ is equivalent to maximizing $\ell(\beta) = \log L(\beta)$.

Log-likelihood

$$\ell(\beta) = \sum_{i=1}^{n} \left[y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right]$$

- Intuition: each observation contributes a reward/penalty depending on how well it is predicted.

How Are the Coefficients Found in Practice?

Important practical point

Unlike linear regression, logistic regression does **not** have a simple closed-form solution for $\hat{\beta}$.

- We solve:

$$\hat{\beta} = \arg\max_{\beta} \ \ell(\beta)$$

- The log-likelihood is smooth and (for standard logistic regression) concave $\Rightarrow$ a unique optimum.
- Numerical algorithms used in software:
  - Gradient ascent / quasi-Newton (e.g., BFGS)
  - Newton–Raphson / IRLS (Iteratively Reweighted Least Squares)

In practice: R / Python / MATLAB compute $\hat{\beta}$ automatically.

Interpreting Logistic Regression Coefficients

- Logistic regression models the **log-odds**:

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta^\top x$$

- Interpretation of β_j:
  - a one-unit increase in x_j changes the log-odds by β_j
  - holding other predictors constant
- Exponentiating gives an **odds ratio**: $\exp(\beta_j) =$ multiplicative change in the odds for a 1-unit increase in x_j

Quick intuition

$\beta_j > 0$ increases odds (and usually probability); $\beta_j < 0$ decreases odds.

Summary

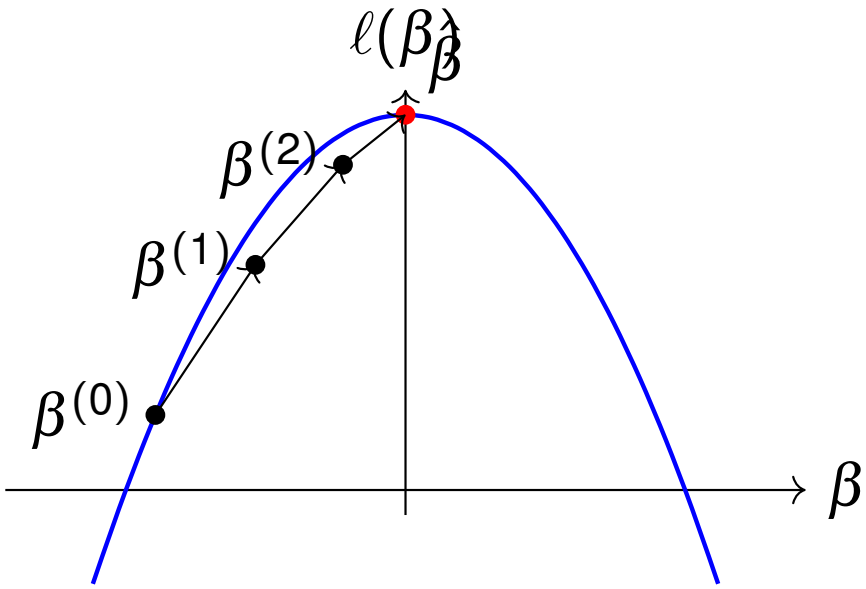
Logistic regression: estimation and meaning

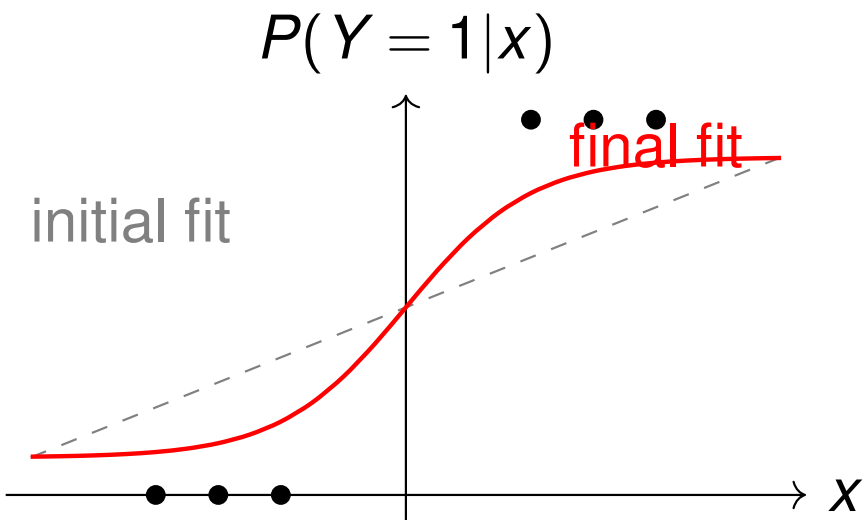
- We model probabilities: $p_i = \mathbb{P}(Y_i = 1 \mid x_i)$.
- Coefficients are estimated by **maximum likelihood**.
- We maximize the log-likelihood:

$$\ell(\beta) = \sum_{i=1}^{n} \Big[y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \Big].$$

- Optimization is numerical (IRLS / Newton methods).
- Coefficients are interpreted via log-odds and odds ratios.

Visual Intuition





Example Estimating the Regression Coefficients

Table 1 shows the coefficient estimates and related information that result from fitting a logistic regression model on the Default data in order to predict the probability of default=Yes using balance.

	Coefficient	Std. error	z-statistic	p-value
`Intercept`	-10.6513	0.3612	-29.5	<0.0001
`balance`	0.0055	0.0002	24.9	<0.0001

For the Default data, estimated coefficients of the logistic regression model that predicts the probability of default using balance. A one-unit increase in balance is associated with an increase in the log odds of default by 0.0055 units.

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Multiple Logistic Regression

By analogy with the extension from simple to MLR , we can generalize (4) as follows, where X=($X_1, \ldots,_p$) are p predictors.

$$\log \frac{\mathrm{p(X)}}{1-\mathrm{p(X)}} = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

Table 2 shows the coefficient estimates that uses balance, income, and student status to predict probability of default.

	Coefficient	Std. error	z-statistic	p-value
`Intercept`	−10.8690	0.4923	−22.08	<0.0001
`balance`	0.0057	0.0002	24.74	<0.0001
`income`	0.0030	0.0082	0.37	0.7115
`student[Yes]`	−0.6468	0.2362	−2.74	0.0062

Student status is encoded as a dummy variable student[Yes], with a value of 1 for a student.

Interpreting Coefficients in Multiple Logistic Regression

- The model describes changes in **log-odds**

$$\log\left(\frac{p(X)}{1-p(X)}\right)$$

Interpretationof a single coefficient β_j

A one-unit increase in X_j changes the log-odds of $Y = 1$ by β_j, **holding all other predictors fixed**.

- This is different from simple logistic regression
- Effects may change sign or significance when new variables are added

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5 Lab: Classifcation Methods

Beyond Binary Logistic Regression

- Logistic regression handles $Y \in \{0, 1\}$
- Many problems involve more than two classes:
  - disease type
  - risk level
  - object category
- We need a model for $Y \in \{1, \ldots, K\}$

Solution

Multinomial (multiclass) logistic regression

Reference Class Idea

- One class is chosen as a baseline (reference)
- Without loss of generality, choose class K
- All other classes are compared to this reference

Why do we need a reference?

- To avoid redundancy
- To ensure model identifiability

Multinomial Logistic Regression Model

For $k = 1, \ldots, K-1$:

$$\Pr(Y = k \mid X = x) = \frac{e^{\beta_{k0} + \beta_k^\top x}}{1 + \sum_{\ell=1}^{K-1} e^{\beta_{\ell 0} + \beta_\ell^\top x}}$$

For the reference class:

$$\Pr(Y = K \mid X = x) = \frac{1}{1 + \sum_{\ell=1}^{K-1} e^{\beta_{\ell 0} + \beta_\ell^\top x}}$$

Key property

Probabilities are positive and sum to 1

Interpretation via Log-Odds

For $k = 1, \ldots, K-1$:

$$\log\left(\frac{\Pr(Y = k \mid X = x)}{\Pr(Y = K \mid X = x)}\right) = \beta_{k0} + \beta_k^\top x$$

Interpretation

- Each class has its own regression equation
- Coefficients compare class k to the reference class
- Effects are interpreted **holding other predictors fixed**

Example

- *Consider an emergency room setting with three possible medical conditions: stroke, drug overdose, and epileptic seizure.*
- *Multinomial logistic regression requires choosing one category as a baseline (reference) class. For example, we may choose stroke or drug overdose as the baseline.*
- *Changing the baseline affects the numerical values of the coefficients, but **does not affect**:*
  - *the predicted probabilities,*
  - *the log-odds between any pair of classes,*
  - *or the overall model fit.*
- *Therefore, coefficient interpretation must always be made relative to the chosen baseline.*

Multinomial Logistic Regression: Softmax Coding

- In softmax coding, **no reference class is chosen**.
- All K classes are treated **symmetrically**.
- Each class k is assigned a linear score:

$$\eta_k(x) = \beta_{k0} + \beta_{k1}x_1 + \cdots + \beta_{kp}x_p$$

Softmax probabilities

$$\Pr(Y = k \mid X = x) = \frac{e^{\eta_k(x)}}{\sum_{\ell=1}^{K} e^{\eta_\ell(x)}}$$

- The denominator ensures probabilities are positive and sum to 1.

Interpretation of Softmax Coefficients

Log-odds between two classes

$$\log\left(\frac{\Pr(Y=k \mid X=x)}{\Pr(Y=k' \mid X=x)}\right) = (\beta_{k0} - \beta_{k'0}) + (\beta_k - \beta_{k'})^\top x$$

- Interpretation is always relative between two classes.
- Individual coefficients β_k have no standalone meaning.
- Only differences between coefficients matter.

Key message

Softmax coding and baseline coding give the same predictions; they are just two equivalent parameterizations.

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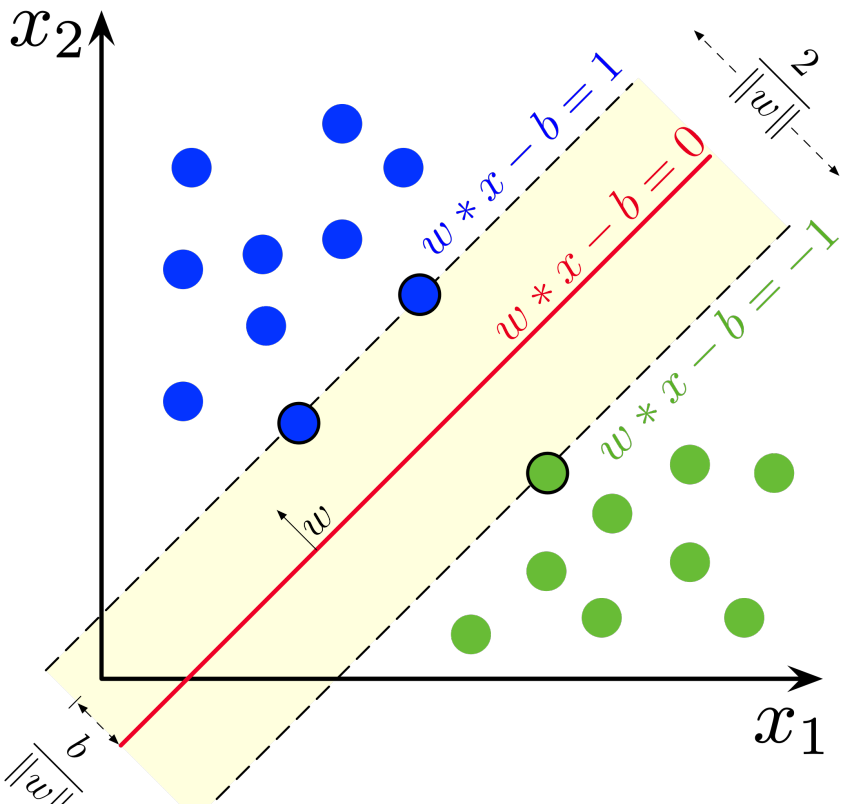
Support Vector Machine (SVM)

Definition

A Support Vector Machine (SVM) is a supervised learning algorithm that finds an optimal hyperplane separating two classes.



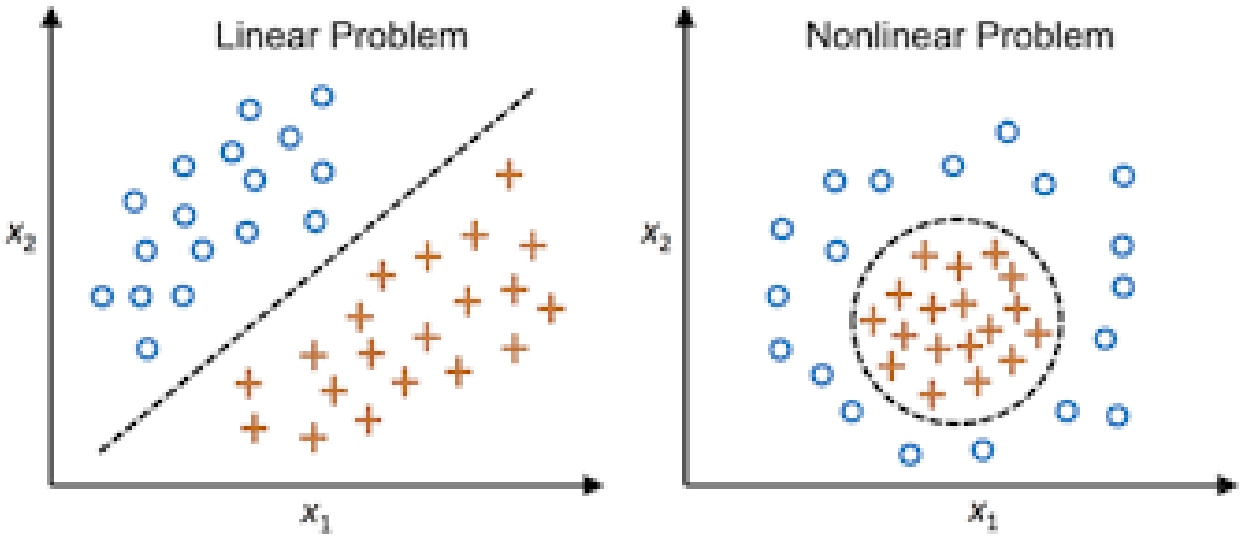
Linearly vs Nonlinearly Separable Data

Linearly separable

- A straight line separates classes
- Simple hyperplane works

Nonlinearly separable

- No straight line can separate classes
- Need feature transformation

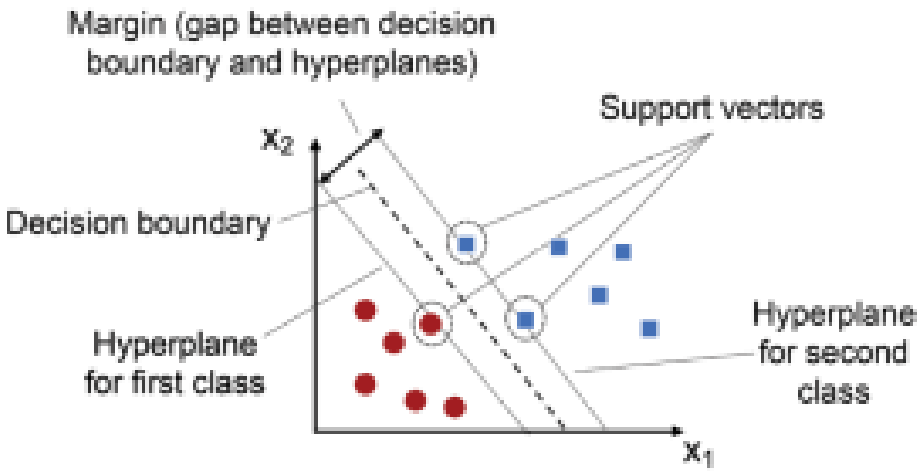


Maximum Margin Principle

Key Idea

Among all separating hyperplanes, SVM chooses the one that maximizes the margin Margin $= \frac{2}{\|\mathbf{w}\|}$.

- Margin = distance to closest points
- Closest points = **Support Vectors**



Hard-Margin SVM (Linearly Separable Data)

Optimization problem

$$\min_{\mathbf{w},b} \quad \frac{1}{2}\|\mathbf{w}\|^2$$

subject to: $y_i(\mathbf{w}^T x_i + b) \geq 1$

Interpretation

- Data are perfectly separable
- No misclassification allowed
- Maximizing margin $\iff$ minimizing $\|\mathbf{w}\|^2$
- Margin width = $\frac{2}{\|\mathbf{w}\|}$

Limitation

Real datasets are rarely perfectly separable.

Why Do We Need Soft Margin?

Problem in practice

- Noise
- Outliers
- Overlapping classes

Idea

Allow violations of the margin constraint using slack variables ξ_i:

$$y_i(\mathbf{w}^T x_i + b) \geq 1 - \xi_i$$

$$\xi_i \geq 0$$

- $\xi_i = 0$ correctly classified outside margin
- $0 < \xi_i < 1$ inside margin
- $\xi_i > 1$ misclassified

Soft-Margin SVM (Primal Form)

$$\min_{\mathbf{w},b,\xi} \quad \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i$$

subject to:

$$y_i(\mathbf{w}^T x_i + b) \geq 1 - \xi_i \quad , \quad \xi_i \geq 0$$

Interpretation

- $\frac{1}{2}\|\mathbf{w}\|^2 \Rightarrow$ maximize margin
- $C\sum\xi_i \Rightarrow$ penalize violations
- C controls trade-off:
  - Small C wider margin, more tolerance
  - Large C stricter classification

Key Intuitions

We want to allow some violations of the margin using slack variables ξ_i.

- $\|w\|^2$ controls the margin size
- ξ_i measures how much point i violates the margin
- C controls how strongly violations are penalized

The constraint

$$y_i(w^\top x_i + b) \geq 1 - \xi_i$$

can be rewritten as

$$\xi_i \geq 1 - y_i(w^\top x_i + b)$$

Therefore ξ_i must satisfy

$$\xi_i \geq \max(0,\ 1 - y_i(w^\top x_i + b))$$

From Slack Variables to Hinge Loss

For fixed (w, b), the optimizer will choose the smallest feasible value of ξ_i. Since we minimize

$$\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i$$

the optimal slack value is

$$\xi_i^\star = \max(0,\ 1 - y_i(w^\top x_i + b))$$

Substituting this into the objective gives

$$\min_{w,b}\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\max(0,\ 1 - y_i(w^\top x_i + b))$$

Interpretation

Let

$$m_i = y_i f(x_i)$$

be the signed margin.

Situation	Margin m_i	Loss
Correct and outside margin	$m_i \geq 1$	0
Inside margin	$0 < m_i < 1$	$1 - m_i$
Misclassified	$m_i \leq 0$	> 1

Key intuition

Hinge loss penalizes margin violations, not only misclassifications.

SVM as Regularized Risk Minimization

The SVM objective becomes

$$\min_{w,b} \underbrace{\sum_{i=1}^{n} \ell_{\text{hinge}}(y_i, f(x_i))}_{\text{regularization}} + \underbrace{\lambda \|w\|^2}_{\text{data fitting}}$$

General ML framework

$$\text{Loss} + \lambda \times \text{Penalty}$$

Interpretation

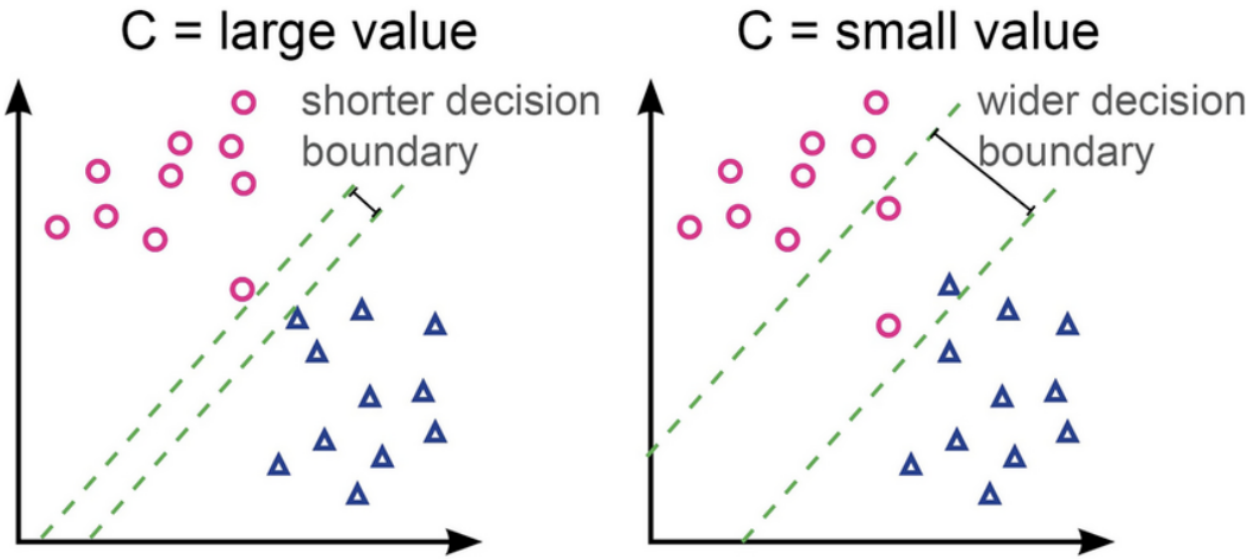
- Loss: penalizes margin violations
- Regularization: encourages a large margin
- Trade-off controlled by C (or λ)

Role of Penality Parameter C

- Large $C \Rightarrow$ small margin low tolerance to errors
- Small $C \Rightarrow$ large margin more tolerance to errors

Bias-Variance Tradeoff

- Large $C \rightarrow$ risk of overfitting
- Small $C \rightarrow$ risk of underfitting



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From Primal to Dual Formulation

Soft-margin SVM (primal):

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \xi_i$$

subject to: $y_i(w^T x_i + b) \geq 1 - \xi_i$

Dual problem

$\max_\alpha \sum_{i=1}^{n} \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle$; $0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i y_i = 0$

Key observation

Data appear only through inner products $\langle x_i, x_j \rangle$

Mathematical Definition of a Kernel

Definition

A function

$$K : \mathscr{X} \times \mathscr{X} \to \mathbb{R}$$

is a **kernel** if there exists a feature map $\phi : \mathscr{X} \to \mathscr{H}$ such that:

$$K(x, x') = \langle \phi(x), \phi(x') \rangle_{\mathscr{H}}$$

Positive Definite Condition

For any finite set $\{x_1, \ldots, x_n\}$, the kernel matrix $K_{ij} = K(x_i, x_j)$ must be symmetric and positive semi-definite:

$$\sum_{i,j} c_i c_j K(x_i, x_j) \geq 0$$

Mercer's Theorem (Intuition)

Mercer's Theorem

If K is continuous, symmetric and positive definite, then there exists a (possibly infinite-dimensional) feature map ϕ such that:

$$K(x, x') = \langle \phi(x), \phi(x') \rangle$$

- We never need to compute $\phi(x)$ explicitly
- Computations happen in input space
- Equivalent to working in a Reproducing Kernel Hilbert Space (RKHS)

Common Kernel Functions

1. Linear Kernel

$$K(x, x') = x^T x'$$

Equivalent to classical linear SVM.

2. Polynomial Kernel

$$K(x, x') = (x^T x' + c)^d$$

- d: degree
- c: offset

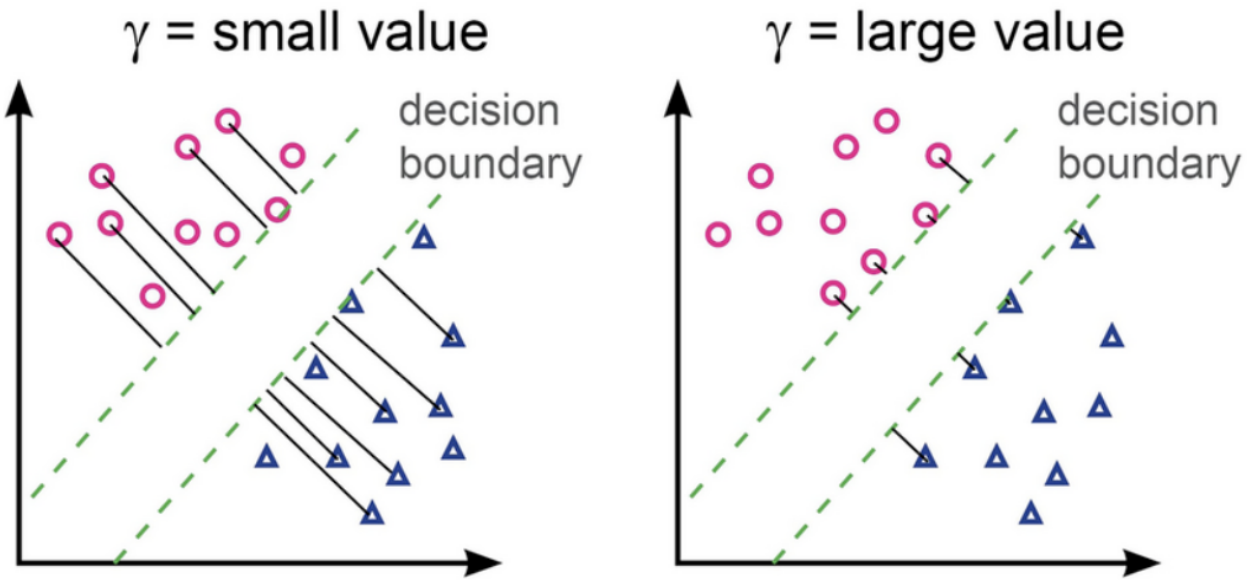
3. Gaussian (RBF) Kernel

$$K(x, x') = \exp(-\gamma \|x - x'\|^2)$$

Understing the RBF Kernel

$$K(x, x') = \exp\left(-\gamma \|x - x'\|^2\right)$$

- Small γ: smoother boundary
- Large γ: complex boundary



Other Valid Kernels

Sigmoid Kernel

$$K(x, x') = \tanh(\kappa x^T x' + c)$$

Related to neural networks.

Laplacian Kernel

$$K(x, x') = \exp(-\gamma \|x - x'\|_1)$$

String / Graph Kernels

Kernels can be defined on:

- Text sequences
- Graphs
- Images
- Structured objects

Designing Custom Kernels

Kernel Engineering

You can define your own kernel as long as it is:

- Symmetric
- Positive semi-definite

Closure Properties

If K_1 and K_2 are valid kernels, then:

- $K = K_1 + K_2$ is valid
- $K = K_1 \cdot K_2$ is valid
- $K = aK_1$ (for $a > 0$) is valid

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Implemention in Python

Main hyperparameters

- `C`: regularization parameter $\rightarrow$ Controls margin vs classification errors Small C = wider margin (more regularization) Large C = fewer errors (less regularization)
- `kernel`: transformation of feature space
  - `"linear"` â linear decision boundary
  - `"poly"` â polynomial boundary
  - `"rbf"` â non-linear Gaussian boundary
- `gamma` (for rbf/poly kernels): Controls how far the influence of a point reaches
- `degree`: degree of polynomial (only if kernel="poly")

SVM Implementation in Python (scikit-learn)

Import

```
from sklearn.svm import SVC
```

Model definition

```
model = SVC(
    C=1.0,
    kernel="rbf",
    gamma="scale",
    degree=3,
    probability=True
)
```

Training and prediction

```
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Using a Custom Kernel in scikit-learn

```
def custom_kernel(X, Y):
    return (X @ Y.T + 1)**2

from sklearn.svm import SVC

model = SVC(kernel=custom_kernel)
model.fit(X_train, y_train)
```

Important

Kernel must produce a positive semi-definite matrix.

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5 Lab: Classifcation Methods

Multi-class SVM

Binary nature of SVM

The standard SVM decision function is binary:

$$f(x) = \mathbf{w}^\top \phi(x) + b \quad \Rightarrow \quad \hat{y} = \mathrm{sign}(f(x)) \in \{-1, +1\}.$$

Multi-class setting

For K classes, we need a rule to combine several binary SVMs into:

$$\hat{y} \in \{1, \ldots, K\}.$$

- Two standard decompositions: **One-vs-Rest (OvR)** and **One-vs-One (OvO)**
- Both are widely used; trade-off is mainly computational and interface / outputs

One-vs-Rest (OvR): K Classifiers

Construction

Train K binary classifiers. For class k:

$$f_k(x) = \mathbf{w}_k^\top \phi(x) + b_k,$$

where positives are class k and negatives are all other classes.

Prediction:

$$\hat{y} = \arg\max_{k \in \{1,\dots,K\}} f_k(x).$$

Pros / Cons

- + Only K models
- + Natural decision scores of shape (n_{samples}, K)
- – Each classifier faces a potentially imbalanced problem (one class vs all)

One-vs-One (OvO): $\frac{K(K-1)}{2}$ Classifiers

Construction

Train one classifier for each pair of classes (k, ℓ):

$$f_{k\ell}(x) = \mathbf{w}_{k\ell}^{\top}\phi(x) + b_{k\ell}.$$

Number of classifiers:

$$\frac{K(K-1)}{2}.$$

Prediction (concept)

Each classifier votes for one of its two classes; final class is obtained by:

$$\hat{y} = \arg\max_{k} \text{votes}(k)$$

(or by aggregating margins / decision values depending on implementation).

What scikit-learn Does (SVC vs LinearSVC)

Key implementation facts

- `SVC` (kernel SVM) trains internally using **OvO**
- Output interface can be formatted as OvR-shaped using: `decision_function_shape="ovr"`
- `LinearSVC` uses OvR directly (trains K models)

```
from sklearn import svm
X = [[0],[1],[2],[3]]
y = [0,1,2,3]
clf = svm.SVC(decision_function_shape="ovo")
clf.fit(X,y)
clf.decision_function_shape = "ovr"
lin = svm.LinearSVC()
lin.fit(X,y)
```

Choosing OvO vs OvR in Practice

Rule of thumb

- If K is small (e.g., $K \leq 10$): OvO is often fine and can be strong.
- If K is large: OvR is typically cheaper (only K models).

Kernel vs Linear

- Kernel SVM (`SVC`): powerful but scaling can be costly with large n
- Linear SVM (`LinearSVC` / `SGDClassifier` with hinge loss): good for high-dimensional features and large datasets

Important practical note

Always scale features (e.g., `StandardScaler`) before SVM.

Outline

1. Classification
2. Logistic Regression
3. SVM
4. Evaluation metrics
5. Lab: Classifcation Methods

Confusion Matrix: the foundation

Binary classification outcomes

	Predicted 0	**Predicted 1**
Actual 0	TN	FP
Actual 1	FN	TP

- TP: true positives (correctly detected positives)
- FP: false positives (false alarms)
- FN: false negatives (missed positives)
- TN: true negatives

Accuracy

Definition

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- Measures overall correctness
- Easy to interpret

Limitation

Accuracy can be misleading when classes are imbalanced.

`sklearn.metrics.accuracy_score`

Precision and Recall

Definitions

$$\text{Precision} = \frac{TP}{TP + FP} \qquad \text{Recall} = \frac{TP}{TP + FN}$$

- **Precision**: How many predicted positives are correct?
- **Recall**: How many actual positives are detected?

Interpretation

- High precision $\Rightarrow$ few false alarms
- High recall $\Rightarrow$ few missed detections

`precision_score, recall_score`

F1-score

Definition

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Harmonic mean of precision and recall
- Penalizes extreme imbalance between the two

When to use F1-score

When both false positives and false negatives matter.

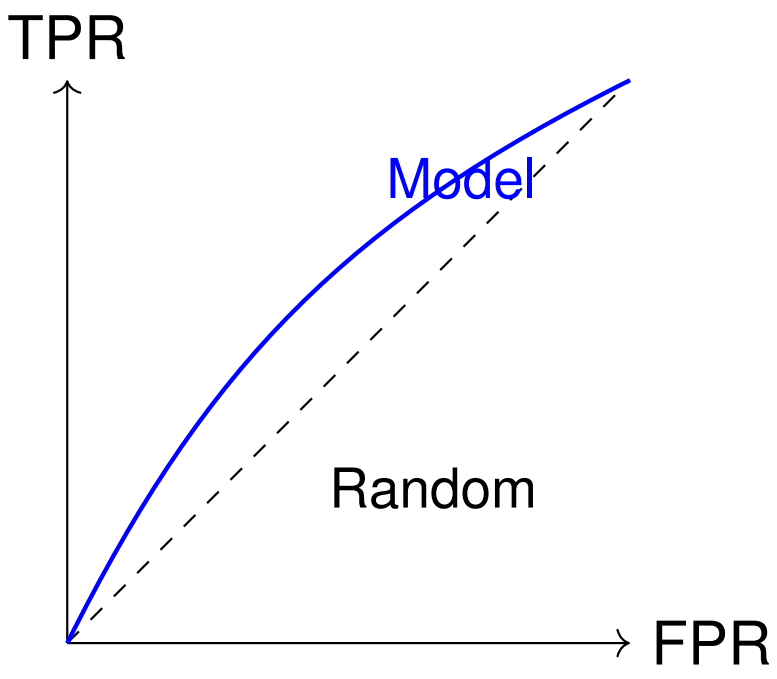
`sklearn.metrics.f1_score`

ROC Curve

Key idea

The ROC curve evaluates performance across **all decision thresholds**.

- TPR = Recall = $\frac{TP}{TP+FN}$
- FPR = $\frac{FP}{FP+TN}$
- Each point = one threshold



AUC: Area Under the Curve

Definition

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR})$$

- Threshold-independent metric
- Probability that a random positive is ranked above a random negative

Interpretation

- AUC = 0.5 (random guessing)
- AUC = 1.0 (perfect classifier)

`sklearn.metrics.roc_auc_score`

Outline

1. Classification
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Lab 1: Social_Network_Ads (Logistic Regression)

Objective

Build a logistic regression classifier to predict whether a user purchases a product (`Purchased` $\in \{0, 1\}$) using two predictors: `Age` and `EstimatedSalary`.

The dataset contains user profiles from a social network advertisement campaign. Each row corresponds to one user.

Instructions

- Open a Python notebook (Jupyter or Google Colab).
- Create one section per step of the lab (Step 1 till 5).
- For each step: include your code, the requested plots/tables, and short answers to the questions.

Deadline: March 24

Step 1: Load & Inspect the Dataset

- Load the CSV file and inspect the table structure
- Identify predictors (`Age`, `EstimatedSalary`) and target (`Purchased`)
- Check missing values and class balance

Questions

- How many observations and variables?
- Is the target balanced?
- Any missing values?

Python hints

`pd.read_csv`, `df.head()`, `df.info()`, `value_counts()`, `isna().sum()`

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Step 2: Visualize the Classes (Separability?)

- Create a scatter plot in feature space: `Age` (x-axis) vs `EstimatedSalary` (y-axis)
- Color points by `Purchased` (0/1)

Questions

- Do the classes appear (approximately) linearly separable?
- What would a linear decision boundary look like here?

Python hints

- `import matplotlib.pyplot as plt`
- `plt.scatter(x, y, c=labels)`
- Color options such as `cmap="bwr"` and transparency `alpha=0.7`
- Axis labels with `plt.xlabel()` and `plt.ylabel()`

Step 3: Fit Logistic Regression (Train/Test)

- Split data into training and test sets
- Fit logistic regression on (`Age`, `EstimatedSalary`)
- Extract coefficients and interpret signs

Questions

- What are the signs of the coefficients?
- Which predictor increases the odds of purchase?

Python hints

- Split the data: `train_test_split` from `sklearn.model_selection`
- Fit the model: `LogisticRegression` from `sklearn.linear_model`
- Access coefficients: `clf.coef_` and `clf.intercept_`

Step 4: Predictions and Model Evaluation

- Use the trained model to predict:
  - class labels (0/1)
  - predicted probabilities
- Evaluate how well the model performs on the test set

Questions

- What is the difference between predicted probabilities and predicted classes?
- How accurate is the model on the test set?
- Is accuracy alone sufficient to evaluate performance?

Python hints

- `clf.predict(X_test)`
- `clf.predict_proba(X_test)`
- `accuracy_score` from `sklearn.metrics`

Step 5: Effect of the Classification Threshold

Logistic regression outputs probabilities in $[0, 1]$. A threshold τ is used to convert probabilities into class labels; We usually fix $\tau = 0.5$, but this is not always optimal

- Recompute confusion matrices for different values of τ

Questions

- What happens if the threshold is lowered (e.g. $\tau = 0.3$)?
- What happens if the threshold is increased (e.g. $\tau = 0.7$)?
- Which types of errors are affected (false positives vs false negatives)?

Python hints

- `clf.predict_proba(X_test)[:,1]`
- Convert to labels: `(proba >= tau).astype(int)`

Step 6: Regularization

Logistic regression estimates coefficients by maximizing the likelihood. However, this can lead to overfitting when:

- predictors are strongly correlated,
- the dataset is small, or we include many features.

Regularized logistic regression maximizes a penalized log-likelihood:

$$\hat{\beta} = \arg\max_{\beta} \Big(\ell(\beta) - \lambda \, \mathscr{P}(\beta) \Big)$$

- $\ell(\beta)$: log-likelihood (fit to the data)
- $\mathscr{P}(\beta)$: penalty on model complexity

Penalty terms

- L2 (Ridge): $\mathscr{P}_{L2}(\beta) = \sum_j \beta_j^2$
- L1 (Lasso): $\mathscr{P}_{L1}(\beta) = \sum_j |\beta_j|$

Step 6: Regularization

- L2 regularization (Ridge)
  - Penalizes the square of coefficients
  - Shrinks coefficients smoothly toward zero
  - Usually keeps all predictors
- L1 regularization (Lasso)
  - Penalizes the absolute value of coefficients
  - Can set some coefficients exactly to zero
  - Performs implicit feature selection

Intuition

- L2:Use all variables, but more cautiously.
- L1: Use only the most important variables.

Questions

- Which regularization would you choose if interpretability is important? Which one would you expect to be more stable?

Step 6: Regularization

In `scikit-learn`, regularization is controlled by the parameter `C`.

- **Large** $C \Rightarrow$ weak regularization (model fits data closely)
- **Small** $C \Rightarrow$ strong regularization (coefficients shrink)

Questions

- Try several values of `C` (e.g. 10, 1, 0.1, 0.01).
- How do the coefficients change as C decreases?
- How does test accuracy change?

Python hints

- `LogisticRegression(penalty="l2", C=C)`
- `LogisticRegression(penalty="l1", solver="liblinear", C=C)`